Neural Network Based Language Models



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One-Hot Encoding

- Before we apply any deep learning techniques, it's important to **encode words into some vectors which can be used as word representation**.
- One such representation technique is **one-hot encoding.**
- In one hot encoding, every word (even symbols) which are part of the given text data are **written in the form of vectors, constituting only of 1 and 0 .**
- Dimension of the **one hot representations are same as size of Vocabulary**

Example

Let's accume ***V*** is the vocabulary which contain 6 words:

$$V = \{\ \text{'This'},\ \text{'is'},\ \text{'data'},\ \text{'a'},\ \text{'science'},\ \text{'course'}\ \}$$

For a given sentence one-hot representation will be:-

"This is a data science course."

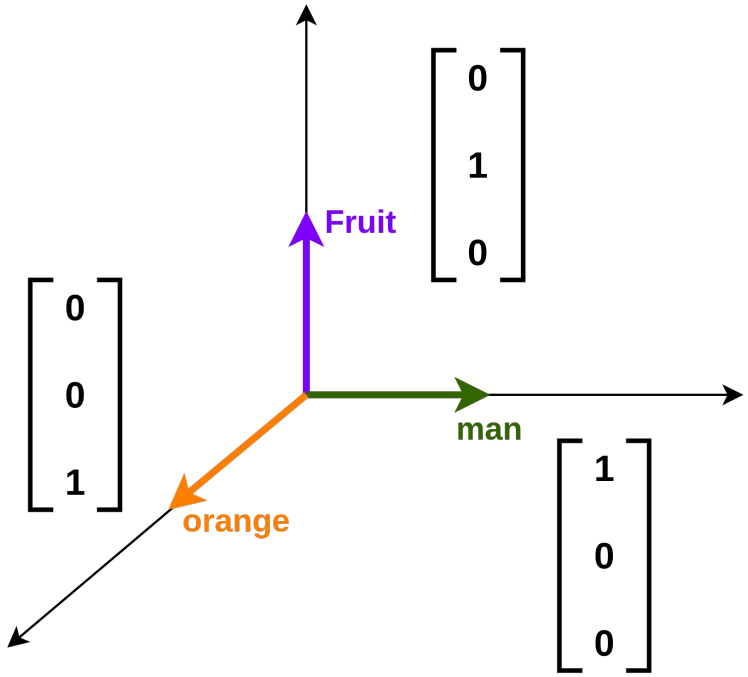
$$\text{"This"} = \begin{bmatrix}1\\0\\0\\0\\0\\0\end{bmatrix} \quad \text{"is"} = \begin{bmatrix}0\\1\\0\\0\\0\\0\end{bmatrix} \quad \text{"a"} = \begin{bmatrix}0\\0\\0\\1\\0\\0\end{bmatrix} \quad \text{"data"} = \begin{bmatrix}0\\0\\1\\0\\0\\0\end{bmatrix} \quad \text{"science"} = \begin{bmatrix}0\\0\\0\\0\\1\\0\end{bmatrix} \quad \text{"course"} = \begin{bmatrix}0\\0\\0\\0\\0\\1\end{bmatrix}$$

Limitations of one-hot encoding

- First major flaw of one hot encoding is that dimension of **one hot vectors depend on vocabulary size.** Usually the Vocabulary size is very large, which result into very large sparse vectors.
- Second limitation is that **one-hot representation does not provide any relative information between two words.**

Limitations of one-hot encoding

Example: The word representations for "**fruit**" and "**orange**" should have more similarity as compared to "**fruit**" and "**man**".



Idea of Word embedding

Suppose there are some **set of features like Gender, Royalty, Age, Food items e.t.c**, which define each word in a Vocabulary :-

number of words in Vocabulary

set of features

word / feature	Man	Woman	King	Queen	. . .	Apple	Orange
Gender	-1	1	-0.95	0.97		0.00	0.01
Royalty	0.01	0.02	0.7	0.8		0.03	0.02
Age	0.53	0.62	0.7	0.69		0.3	0.2
. . .							
Food Items	0.01	0.00	0.03	0.01		0.95	0.97

- Darker shades and lighter shades indicates positive feature values and negative feature values respectively.
- For example, from the given table we can say that “gender” is an important feature for word like “Man, Woman, King, Queen” and does not have any importance for word like “Apple, Oranges”.
- Is there a way to **learn such a feature map ?**

Embedding Matrix

The **feature map discussed earlier is also called as embedding matrix** (***E***). The use of embedding matrix is to obtain meaningful word embedding using one-hot representations.

$$e_{d\times 1} = E_{d\times|V|} \times o_{|V|\times 1}$$

This represent a one hot vector of size equal to size of Vocabulary (|V|)

$$e_{d\times 1} = E_{d\times |V|} \times o_{|V|\times 1}$$

Word embedding which represent a word (or any token) of size d

Learnable Embedding matrix (***E***) with number of columns as same as size of vocabulary (**|V|**) and and number of rows as number of features (***d***). For example :-

|V|
number of words in Vocabulary

d
set of features

word / feature	Man	Woman	King	Queen	. . .	Apple	Orange
Gender	-1	1	-0.95	0.97		0.00	0.01
Royalty	0.01	0.02	0.7	0.8		0.03	0.02
Age	0.53	0.62	0.7	0.69		0.3	0.2
. . .							
Food Items	0.01	0.00	0.03	0.01		0.95	0.97

Learning Embedding matrix (E)

- Let's assume the size of Vocabulary ($|V|$) is 10,000. To learn the embedding matrix E, given the training corpus, **we can try to solve a word prediction problem.**
- For each sentences we can try to predict the target word, for example:-
- Given a sentence ***"This is a data science course"***, we can try to predict the target word which is ***"course"***.
- This types of problems helps the embedding matrix to learn relative information between words due to which we can get useful word embedding.

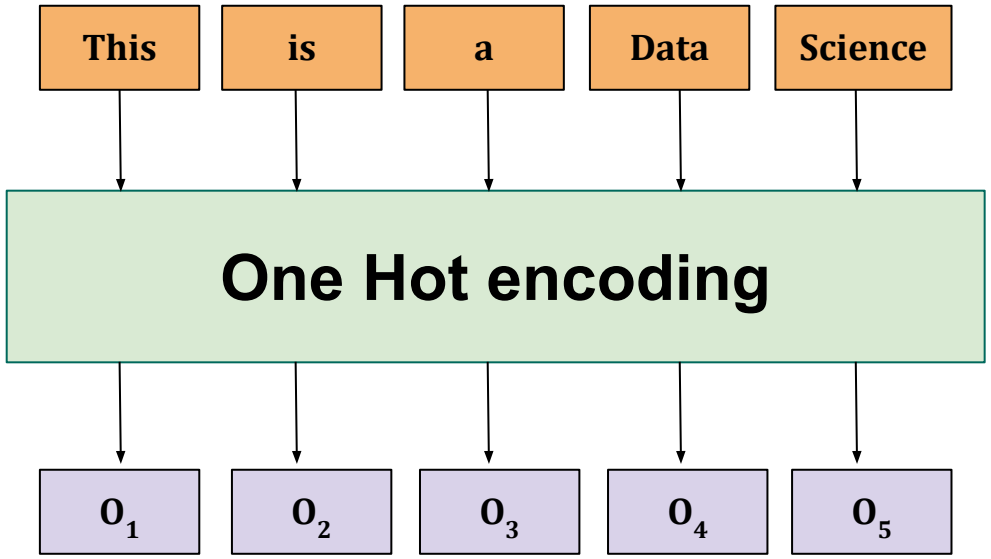
Word2Vec

Word2Vec method is used to construct meaningful word embeddings.

Example:-

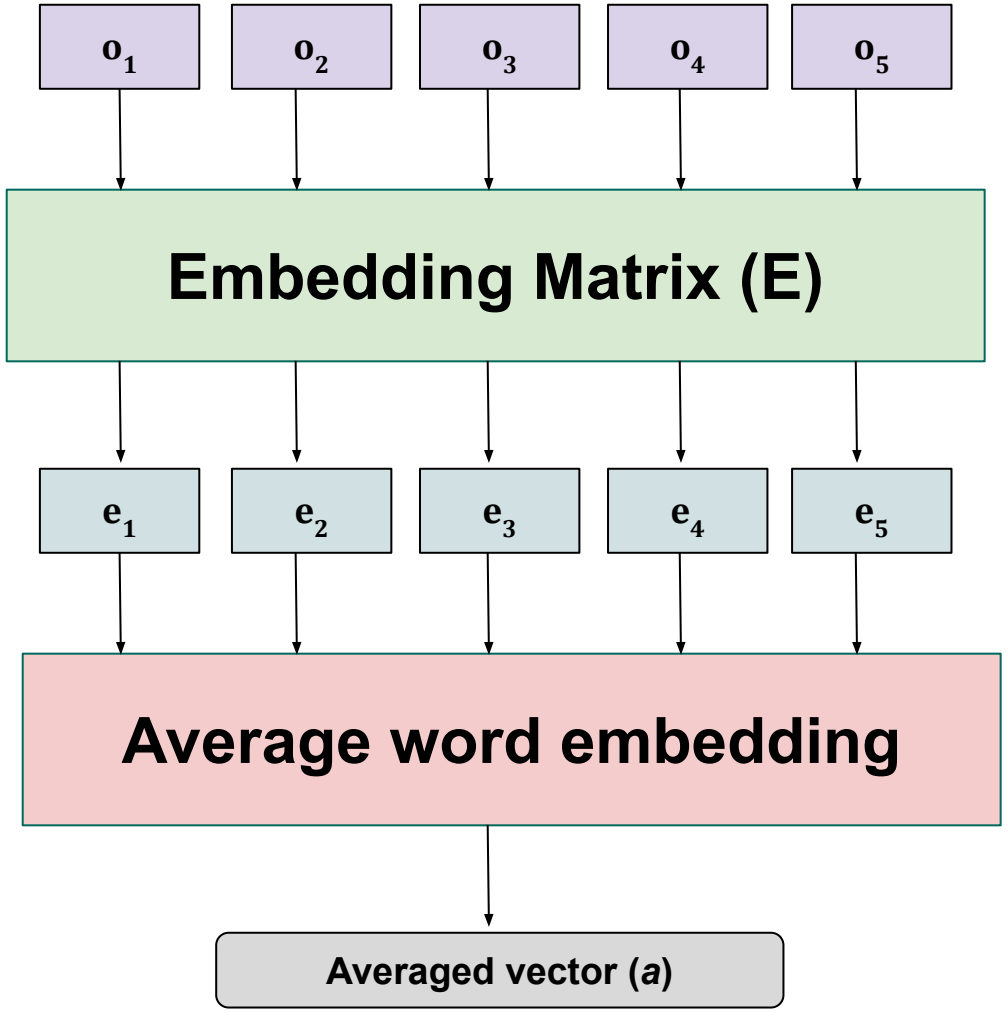


Learning Embedding Matrix (E) - Word2Vec



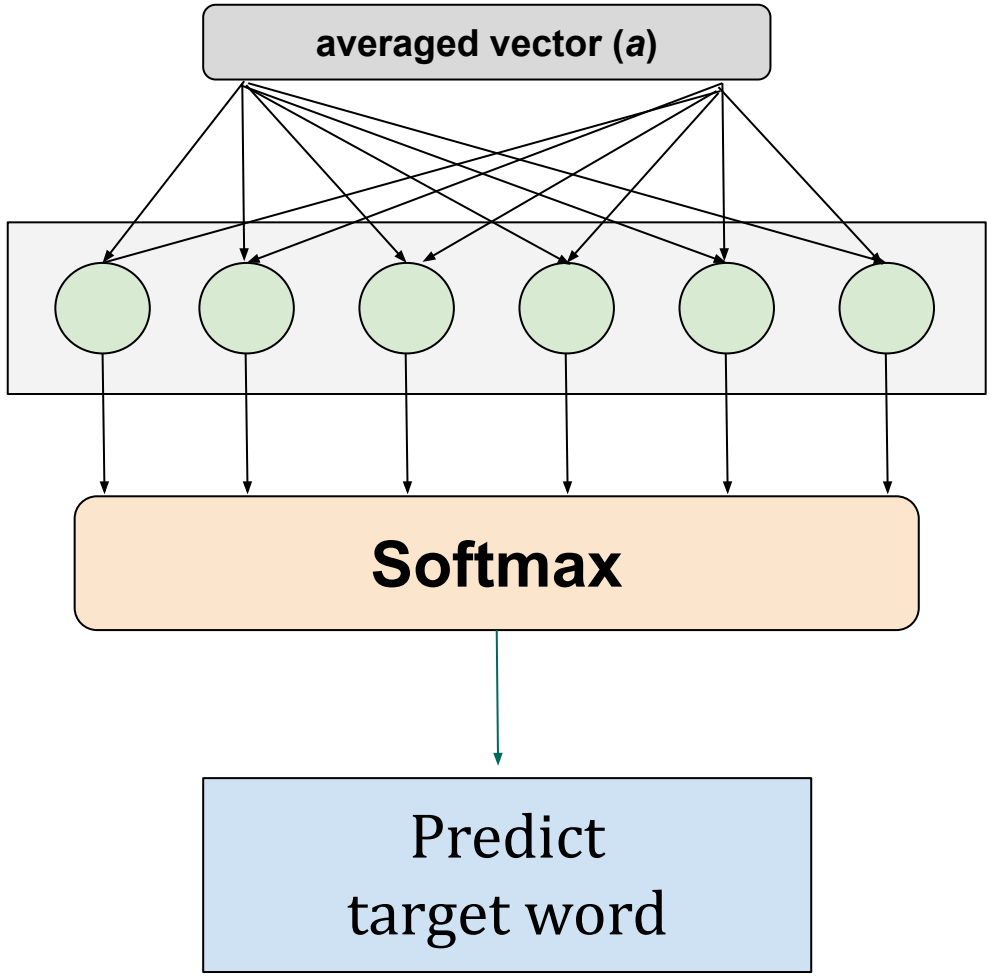
- The first step will be to obtain the one-hot vectors for each of the words.
- The size each vector will be 10,000 (based on the size of the vocabulary)

Learning Embedding Matrix (E) - Word2Vec



- After we obtain the **one-hot vectors (o_i), we multiply those vectors with randomly initialized embedding matrix (E) to get word embeddings (e_i) of size d.**
- Next these **embeddings are average out to get a single vector *a* of size d.**

Learning Embedding Matrix (E) - Word2Vec



- Final step is to **pass the averaged vector (a) through a linear layer and perform a softmax across all the word in the vocabulary**, to predict target word **"course"**.
- We use **backpropagation to learn embedding matrix (E)**

Limitation of Word2Vec

- The vocabulary size can be very large (can be in millions !). This can increase the softmax computations. To avoid this we can use **Negative Sampling** and **Hierarchical Softmax.**
- The **averaging all the context vector to predict a target word** is not a good strategy. For example :-

